

An Efficient Identity-Based Blind Signature Scheme from SM9

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Abstract. We present an efficient identity-based blind signature scheme built upon the SM9 Chinese national cryptographic standard. Our scheme achieves weak one-more unforgeability under the q -SDH assumption, and computational blindness in the random oracle model (ROM), both proved rigorously. The weak security model, while restricting adversaries from querying signatures on the target identity, remains sufficient for critical applications including electronic cash, anonymous credentials, and privacy-preserving voting systems. We provide rigorous security proofs and demonstrate significant performance improvements over existing SM9-based blind signature schemes, eliminating all online pairing computations through pre-computation. Our construction maintains full compatibility with the SM9 standard, facilitating deployment in certificate-free environments.

Keywords. Blind signature, SM9, Identity-based cryptography, Weak one-more unforgeability, Computational blindness, q -SDH assumption, Random oracle model, Electronic cash.

1 Introduction

Blind signatures, introduced by Chaum in 1982 [1], enable a signer to digitally sign a message without learning its content. This cryptographic primitive has become fundamental to privacy-preserving protocols, particularly in electronic cash (e-cash) systems where users require anonymous yet authenticated transactions. Identity-Based Blind Signatures were first proposed by Zhang and Kim [10]. Cao et al. [11] improved efficiency but lacked formal security proofs. Recently, several identity-based blind signature schemes from SM9 [12,13,14,15] have been proposed, but they either provide informal security arguments or require strong assumptions and additional cryptographic components.

The SM9 identity-based cryptographic standard [6], promulgated by the State Cryptography Administration of China and adopted as ISO/IEC 14888-3:2018, offers significant advantages over traditional PKI-based systems by eliminating certificate management overhead. Cheng [8] and Lai et al. [9] analyzed SM9 key agreement and encryption, establishing security under Gap- q -BCAA1. Our work extends this line by proving blind signature security under the weaker q -SDH assumption in the weakOMU model and computational blindness in the ROM.

Existing SM9-based blind signature schemes face several challenges:

- **Security models:** Most schemes claim security under the standard one-more unforgeability (OMU) model but provide incomplete or flawed proofs, particularly when handling concurrent signing sessions on the target identity.
- **Strong assumptions:** Rigorous proofs often require Gap-type assumptions (e.g., Gap- q -BCAA1) which are stronger than standard computational assumptions.
- **Performance overhead:** Existing constructions involve multiple online pairing computations, limiting their applicability in resource-constrained environments.

Pointcheval and Stern [2,3] formalized OMU for blind signatures. We adopt the *weak one-more unforgeability* (weak OMU) model, which restricts adversaries from querying signatures on the target identity. While this represents a relaxation of the standard OMU model, we argue that weak OMU is sufficient and natural for the primary applications of blind signatures.

Electronic Cash (E-Cash). In Chaum’s e-cash protocol [1], a user withdraws a coin by obtaining a blind signature from the bank on a random serial number. The bank never signs the same serial number twice for the same user account—each withdrawal creates a fresh coin. The weak OMU model precisely captures this scenario: the adversary (malicious user) cannot request multiple signatures on the target coin (the one being forged), as the bank’s policy inherently prevents this.

Anonymous Credentials. In attribute-based credential systems, a user obtains a blind signature on their attributes from an issuer. The issuer signs each credential request exactly once per user. The weak OMU model ensures that users cannot forge additional credentials with the same attributes without interacting with the issuer again.

Privacy-Preserving E-Voting. In blind signature-based voting protocols, the election authority blindly signs each voter’s ballot exactly once (enforced by voter registration). The weak OMU model guarantees that voters cannot forge additional signed ballots for the same election, preventing ballot stuffing attacks.

1.1 Our Contributions

We present the first SM9-based blind signature scheme with:

1. **Rigorous security proofs:** Weak OMU under the standard q -SDH assumption (without Gap assumptions) and computational blindness in the random oracle model, achieved by introducing a user-side multiplicative blinding factor α that hides the session randomness from the signer.
2. **Optimal efficiency:** Zero online pairings (one pre-computable pairing), three exponentiations in $\mathbb{G}_T$, and two scalar multiplications in $\mathbb{G}_1$, achieving approximately 61–82% latency reduction compared to prior work.

3. **Full SM9 compatibility:** Signatures are standard SM9 signatures, verifiable by any SM9-compliant implementation.
4. **Practical applicability:** Detailed analysis of the weak OMU model's sufficiency for e-cash, credentials, and voting systems.

1.2 Organization

Section 2 reviews cryptographic preliminaries. Section 3 formalizes the security models. Section 4 presents our scheme. Section 5 provides security proofs. Section 6 analyzes performance. Section 7 extends to partial blind signatures. Section 8 concludes.

2 Preliminaries

2.1 Bilinear Pairings

Let $\mathbb{G}_1$, $\mathbb{G}_2$, and $\mathbb{G}_T$ be cyclic groups of prime order q , with generators $P_1 \in \mathbb{G}_1$, $P_2 \in \mathbb{G}_2$, and $g \in \mathbb{G}_T$. A bilinear pairing is a map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ satisfying:

- **Bilinearity:** $e([a]P_1, [b]P_2) = e(P_1, P_2)^{ab}$ for all $a, b \in \mathbb{Z}_q$.
- **Non-degeneracy:** $e(P_1, P_2) \neq 1_{\mathbb{G}_T}$.
- **Computability:** e is efficiently computable.

We use Type-3 pairings where no efficient isomorphism between $\mathbb{G}_1$ and $\mathbb{G}_2$ is known.

2.2 Computational Assumptions

Definition 1 (q -Strong Diffie-Hellman (q -SDH)). Given $(P_1, [s]P_1, [s^2]P_1, \dots, [s^q]P_1, P_2, [s]P_2)$ where $s \in \mathbb{Z}_q^*$ is unknown, the q -SDH problem is to compute $(c, [1/(s+c)]P_1)$ for some $c \in \mathbb{Z}_q^*$.

The q -SDH assumption states that for all PPT adversaries $\mathcal{A}$,

$$\text{Adv}_{\mathcal{A}}^{q\text{-SDH}}(\lambda) := \Pr[\mathcal{A}(P_1, [s]P_1, \dots, [s^q]P_1, P_2, [s]P_2) = (c, [1/(s+c)]P_1)] \leq \text{negl}(\lambda).$$

2.3 SM9 Identity-Based Signature

The SM9 signature scheme [6] consists of the following algorithms.

Setup. The Key Generation Center (KGC) selects $s \in \mathbb{Z}_q^*$ as the master secret key and publishes the master public key $P_{\text{pub}} = [s]P_2$. System parameters include $(P_1, P_2, P_{\text{pub}}, H_1, H_2)$ where $H_1 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$ and $H_2 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$ are cryptographic hash functions.

Extract. For identity ID, compute $\tau = H_1(\text{ID} \parallel \text{hid}) \in \mathbb{Z}_q^*$ and return the private key

$$d_{\text{ID}} = \left[\frac{s}{s + \tau} \right] P_1 \in \mathbb{G}_1.$$

Sign. To sign message M with private key d_{ID} :

1. Choose $k \in \mathbb{Z}_q^*$ uniformly at random.
2. Compute $w = e(P_1, P_{\text{pub}})^k = g^k \in \mathbb{G}_T$.
3. Compute $h = H_2(M \| w) \in \mathbb{Z}_q^*$.
4. Compute $S = [k - h] d_{\text{ID}} \in \mathbb{G}_1$.
5. Output signature $\sigma = (h, S)$.

Verify. Given $(\text{ID}, M, \sigma = (h, S))$:

1. Compute $\tau = H_1(\text{ID} \| \text{hid})$.
2. Compute $w' = g^h \cdot e(S, [\tau]P_2 + P_{\text{pub}})$ where $g = e(P_1, P_{\text{pub}})$.
3. Accept iff $h = H_2(M \| w')$.

Correctness.

$$\begin{aligned}
 w' &= g^h \cdot e([k - h]d_{\text{ID}}, [\tau]P_2 + P_{\text{pub}}) \\
 &= g^h \cdot e\left([k - h] \cdot \left[\frac{s}{s + \tau}\right] P_1, [s + \tau]P_2\right) \\
 &= g^h \cdot e([k - h]P_1, [s]P_2) \\
 &= g^h \cdot g^{k - h} = g^k = w. \quad \checkmark
 \end{aligned}$$

3 Security Models

3.1 Syntax of Blind Signatures

A blind signature scheme consists of four algorithms and an interactive protocol:

- $\text{Setup}(1^\lambda) \rightarrow (\text{mpk}, \text{msk})$: Generates master public/secret keys.
- $\text{Extract}(\text{msk}, \text{ID}) \rightarrow d_{\text{ID}}$: Extracts private key for identity ID .
- $\langle \text{User}(M), \text{Signer}(d_{\text{ID}}) \rangle \rightarrow (\sigma, \perp)$: Interactive signing protocol where the user obtains signature σ on message M and the signer learns nothing about M or σ .
- $\text{Verify}(\text{mpk}, \text{ID}, M, \sigma) \rightarrow \{0, 1\}$: Verifies signature σ on message M under identity ID .

3.2 Weak One-More Unforgeability

Definition 2 (Weak One-More Unforgeability). A blind signature scheme is weakly one-more unforgeable if for all PPT adversaries $\mathcal{A}$, the advantage

$$\text{Adv}_{\mathcal{A}}^{\text{weak-OMU}}(\lambda) := \Pr\left[\text{Exp}_{\mathcal{A}}^{\text{weak-OMU}}(\lambda) = 1\right]$$

is negligible, where the experiment is defined as follows.

Experiment $\text{Exp}_{\mathcal{A}}^{\text{weak-OMU}}(\lambda)$:

1. $(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda)$.
2. $(\text{ID}^*, \{(M_i, \sigma_i)\}_{i=1}^n) \leftarrow \mathcal{A}^{\mathcal{O}^{\text{Ext}}, \mathcal{O}^{\text{Sign}}, H_1, H_2}(\text{mpk})$.
3. If ID^* was queried to $\mathcal{O}^{\text{Ext}}$: **return** 0.
4. If $(\text{ID}^*, \cdot)$ was queried to $\mathcal{O}^{\text{Sign}}$: **return** 0. (WeakOMU restriction)
5. If $n < 1$: **return** 0.
6. If $\exists i \neq j : M_i = M_j$: **return** 0.
7. If $\exists i : \text{Verify}(\text{mpk}, \text{ID}^*, M_i, \sigma_i) = 0$: **return** 0.
8. **return** 1.

Here, $\mathcal{O}^{\text{Ext}}(\text{ID})$ returns d_{ID} , and $\mathcal{O}^{\text{Sign}}(\text{ID}, \cdot)$ engages in the signing protocol as the signer with identity ID .

Remark 1 (Comparison with StandardOMU). The standard OMU model allows $\mathcal{A}$ to query $\mathcal{O}^{\text{Sign}}(\text{ID}^*, \cdot)$ up to ℓ times and requires $n > \ell$ forgeries. The weak OMU model sets $\ell = 0$ by disallowing any signing queries on ID^* , thus requiring $n \geq 1$ (at least one forgery without any legitimate signature). This simplification is natural for applications where the signer's policy prevents multiple signatures on the same message for the target identity.

3.3 Computational Blindness

Definition 3 (Computational Blindness). A blind signature scheme has computational blindness in the random oracle model if for all PPT adversaries $\mathcal{A}$,

$$\text{Adv}_{\mathcal{A}}^{\text{blind}}(\lambda) := \left| \Pr[\text{Exp}_{\mathcal{A},0}^{\text{blind}}(\lambda) = 1] - \Pr[\text{Exp}_{\mathcal{A},1}^{\text{blind}}(\lambda) = 1] \right| \leq \text{negl}(\lambda),$$

where the experiment $\text{Exp}_{\mathcal{A},b}^{\text{blind}}(\lambda)$ is:

Experiment $\text{Exp}_{\mathcal{A},b}^{\text{blind}}(\lambda)$:

1. $(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda)$.
2. $\mathcal{A}(\text{mpk})$ outputs $(M_0, M_1, \text{ID}^*, \text{state})$.
3. Run two concurrent signing sessions with $\mathcal{A}$ (acting as signer) and two user instances running on messages M_b (session 0) and M_{1-b} (session 1), using private key d_{ID^*} .
4. $\mathcal{A}$ outputs a bit b' .
5. **return** $[b' = b]$.

4 Our Construction

4.1 Scheme Description

Setup. Run SM9 setup: select $s \in \mathbb{Z}_q^*$, set $P_{\text{pub}} = [s]P_2$, and publish $\text{mpk} = (P_1, P_2, P_{\text{pub}}, H_1, H_2)$. Pre-compute $g = e(P_1, P_{\text{pub}})$.

Extract. For identity ID, compute $\tau = H_1(\text{ID} \parallel \text{hid})$ and return $d_{\text{ID}} = [s/(s + \tau)] P_1$.

Blind Signing Protocol. User (with message M) and Signer (with private key d_A for identity A) execute the following protocol. The key design principle is that the user introduces a secret multiplicative blinding factor $\alpha \in \mathbb{Z}_q^*$, which hides the true session randomness $\tilde{w}$ from the signer in the random oracle model, achieving computational blindness.

1. **User $\rightarrow$ Signer:** Choose $k_1 \xleftarrow{\$} \mathbb{Z}_q^*$. Compute $w_1 = g^{k_1} \in \mathbb{G}_T$. Send w_1 .
2. **Signer $\rightarrow$ User:** Choose $k_2 \xleftarrow{\$} \mathbb{Z}_q^*$. Compute $w_{\text{raw}} = w_1^{k_2} = g^{k_1 k_2} \in \mathbb{G}_T$. Send w_{raw} .
3. **User (local blinding):** Choose $\alpha \xleftarrow{\$} \mathbb{Z}_q^*$ (kept secret from the signer). Compute $\tilde{w} = w_{\text{raw}}^\alpha = g^{k_1 k_2 \alpha} \in \mathbb{G}_T$. Compute $h = H_2(M \parallel \tilde{w}) \pmod{q}$. Compute $h'' = (k_1 \alpha)^{-1} \cdot h \pmod{q}$. Send h'' to Signer.
4. **Signer $\rightarrow$ User:** Compute $l = k_2 - h'' \pmod{q}$. Compute $S' = [l] d_A \in \mathbb{G}_1$. Send S' .
5. **User (local unblinding):** Compute $S = [k_1 \alpha] S' \in \mathbb{G}_1$. Output signature $\sigma = (h, S)$.

Verify. Run standard SM9 verification on $(\text{ID} = A, M, \sigma = (h, S))$.

4.2 Correctness

Proposition 1 (Correctness). *The output $\sigma = (h, S)$ of the blind signing protocol is a valid SM9 signature on M under identity A .*

Proof. Let $k = k_1 \alpha k_2$.

$$\begin{aligned} S &= [k_1 \alpha] S' = [k_1 \alpha] [k_2 - h''] d_A \\ &= [k_1 \alpha k_2 - k_1 \alpha \cdot (k_1 \alpha)^{-1} h] d_A \\ &= [k_1 \alpha k_2 - h] d_A = [k - h] d_A. \end{aligned}$$

Since $\tilde{w} = w_{\text{raw}}^\alpha = g^{k_1 k_2 \alpha} = g^k$ and $h = H_2(M \parallel \tilde{w}) = H_2(M \parallel g^k)$, the pair (h, S) satisfies the SM9 verification equation. $\square$

4.3 Efficiency

Computation Costs.

- **Pre-computation (one-time):** 1 pairing to compute $g = e(P_1, P_{\text{pub}})$.
- **User (online):** 2 exponentiations in $\mathbb{G}_T$ ($w_1 = g^{k_1}$, $\tilde{w} = w_{\text{raw}}^\alpha$), 1 scalar multiplication in $\mathbb{G}_1$ ($S = [k_1 \alpha] S'$), 2 modular operations.
- **Signer (online):** 1 exponentiation in $\mathbb{G}_T$ ($w_{\text{raw}} = w_1^{k_2}$), 1 scalar multiplication in $\mathbb{G}_1$ ($S' = [l] d_A$), 2 modular operations.
- **Total online:** 0 pairings, 3 exponentiations in $\mathbb{G}_T$, 2 scalar multiplications in $\mathbb{G}_1$.

Communication Costs. Four messages: $w_1 \in \mathbb{G}_T$ (512 bits), $w_{\text{raw}} \in \mathbb{G}_T$ (512 bits), $h'' \in \mathbb{Z}_q$ (256 bits), $S' \in \mathbb{G}_1$ (256 bits, compressed). Total: ≈ 1536 bits.

5 Security Analysis

5.1 Weak One-More Unforgeability

Theorem 1 (Weak OMU under q -SDH). *Let H_1, H_2 be modeled as random oracles. If the q -SDH assumption holds, then the scheme is weakly one-more unforgeable. Concretely, for any PPT adversary $\mathcal{A}$ making at most q_{H_1} queries to H_1 , q_{H_2} queries to H_2 , q_e extraction queries, and no signing queries on the target identity ID^* , there exists a PPT algorithm $\mathcal{B}$ such that*

$$\text{Adv}_{\mathcal{A}}^{\text{weak-OMU}}(\lambda) \leq q_{H_2} \sqrt{q_{H_1} \cdot \text{Adv}_{\mathcal{B}}^{q\text{-SDH}}(\lambda)} + q_{H_2} \sqrt{\frac{3}{2q}} + \frac{1}{q_{H_1}}.$$

Proof. We construct a PPT algorithm $\mathcal{B}$ that uses $\mathcal{A}$ to solve the q -SDH problem.

Receiving the q -SDH Challenge. $\mathcal{B}$ receives a q -SDH instance

$$\mathcal{I} = (P_1, [s]P_1, [s^2]P_1, \dots, [s^q]P_1, P_2, [s]P_2),$$

where $s \in \mathbb{Z}_q^*$ is unknown, and must output $(c, [1/(s+c)]P_1)$ for some $c \in \mathbb{Z}_q^*$.

Setup Phase. $\mathcal{B}$ sets $P_{\text{pub}} = [s]P_2$ (from the challenge) and computes $g = e(P_1, P_{\text{pub}}) = e(P_1, P_2)^s$. $\mathcal{B}$ sends $\text{mpk} = (P_1, P_2, P_{\text{pub}}, H_1, H_2)$ to $\mathcal{A}$.

Guessing the Target Identity. $\mathcal{B}$ guesses an index $j^* \xleftarrow{\$} [q_{H_1}]$, anticipating that the j^* -th query to H_1 corresponds to ID^* .

Simulating H_1 Queries. $\mathcal{B}$ maintains a list L_{H_1} . For the j -th distinct query $H_1(\text{ID}_j \| \text{hid})$:

- **If $j = j^*$:** Choose $\tau^* \xleftarrow{\$} \mathbb{Z}_q^*$. Store (ID_j, τ^*) and return τ^* . ($\mathcal{B}$ does not know $d_{\text{ID}_j^*} = [s/(s+\tau^*)]P_1$.)
- **If $j \neq j^*$:** Choose $\tau_j \xleftarrow{\$} \mathbb{Z}_q^*$. Compute d_{ID_j} via the Boneh–Boyen polynomial technique [7]: define

$$F_j(s) := \sum_{i=0}^{q-1} (-\tau_j)^{q-1-i} s^i \in \mathbb{Z}_q[s].$$

One verifies $(s + \tau_j)F_j(s) = s^q - \tau_j^q \equiv s - \tau_j \pmod{q}$ (Fermat), so

$$\frac{s}{s + \tau_j} = \frac{1}{2}(F_j(s) + 1), \quad d_{\text{ID}_j} = \frac{1}{2} \left(\sum_{i=0}^{q-1} (-\tau_j)^{q-1-i} [s^i]P_1 + P_1 \right).$$

Store $(\text{ID}_j, \tau_j, d_{\text{ID}_j})$ and return τ_j .

Simulating H_2 Queries. For each query $H_2(M\|w)$: return a stored response if available; otherwise choose $h \xleftarrow{\$} \mathbb{Z}_q^*$, store (M, w, h) , and return h .

Simulating $\mathcal{O}^{\text{Ext}}$ Queries. For $\mathcal{O}^{\text{Ext}}(\text{ID}_j)$: trigger H_1 if needed; abort if $j = j^*$; otherwise return d_{ID_j} .

Simulating $\mathcal{O}^{\text{Sign}}$ Queries. For $\mathcal{O}^{\text{Sign}}(\text{ID}_j, \cdot)$: trigger H_1 if needed; abort if $j = j^*$ (never reached by weak OMU); otherwise execute the signing protocol honestly using d_{ID_j} :

1. Receive $w_1 \in \mathbb{G}_T$ from $\mathcal{A}$.
2. Choose $k_2 \xleftarrow{\$} \mathbb{Z}_q^*$; compute $w_{\text{raw}} = w_1^{k_2}$; send w_{raw} .
3. Receive $h'' \in \mathbb{Z}_q$ from $\mathcal{A}$.
4. Compute $l = k_2 - h''$, $S' = [l] d_{\text{ID}_j}$; send S' .

Key Observation. In the weak OMU model, $\mathcal{A}$ is not allowed to query $\mathcal{O}^{\text{Sign}}(\text{ID}^*, \cdot)$. Therefore $\mathcal{B}$ never needs to simulate a signing session for ID^* , avoiding the need to know d_{ID^*} .

Forgery Phase. Suppose $\mathcal{A}$ outputs $n \geq 1$ valid forgeries $\{(M_i, \sigma_i = (h_i, S_i))\}_{i=1}^n$ for ID^* , where ID^* was not queried to $\mathcal{O}^{\text{Ext}}$ or $\mathcal{O}^{\text{Sign}}$, all M_i are distinct, and all σ_i are valid SM9 signatures.

$\mathcal{B}$ guesses a target forgery index $i^* \xleftarrow{\$} [n]$; since $n \leq q_{H_2}$, this is correct with probability $\geq 1/q_{H_2}$.

Checking the Guess. $\mathcal{B}$ checks whether ID^* corresponds to the j^* -th H_1 query (probability $1/q_{H_1}$). If not, $\mathcal{B}$ aborts.

Applying the General Forking Lemma. $\mathcal{B}$ applies the General Forking Lemma [5] to $\mathcal{A}$ with respect to the H_2 random oracle, targeting the H_2 query associated with $(M_{i^*}, \tilde{w}_{i^*})$. The Forking Lemma guarantees that with probability at least

$$p_{\text{fork}} \geq \frac{\epsilon^2}{q_{H_2}} - \frac{1}{2q},$$

$\mathcal{B}$ obtains two accepting transcripts with different H_2 responses, yielding two valid SM9 signatures on M_{i^*} under ID^* :

$$\sigma_{i^*} = (h_{i^*}, S_{i^*}), \tag{1}$$

$$\sigma'_{i^*} = (h'_{i^*}, S'_{i^*}), \tag{2}$$

where $h_{i^*} \neq h'_{i^*}$ and both share the same $\tilde{w}_{i^*} = g^{k_{i^*}}$.

Extracting the Private Key. Both signatures satisfy:

$$S_{i^*} = [k_{i^*} - h_{i^*}] d_{\text{ID}^*}, \quad (3)$$

$$S'_{i^*} = [k_{i^*} - h'_{i^*}] d_{\text{ID}^*}. \quad (4)$$

Subtracting (4) from (3):

$$S_{i^*} - S'_{i^*} = [h'_{i^*} - h_{i^*}] d_{\text{ID}^*}.$$

Since $\Delta := h'_{i^*} - h_{i^*} \neq 0$:

$$d_{\text{ID}^*} = [\Delta^{-1}] (S_{i^*} - S'_{i^*}) = \left[\frac{s}{s + \tau^*} \right] P_1.$$

Extracting the q -SDH Solution. From $d_{\text{ID}^*} = [s/(s + \tau^*)] P_1$:

$$\left[\frac{1}{s + \tau^*} \right] P_1 = [(\tau^*)^{-1}] (P_1 - d_{\text{ID}^*}).$$

$\mathcal{B}$ outputs the q -SDH solution $(\tau^*, [(\tau^*)^{-1}](P_1 - d_{\text{ID}^*}))$. $\checkmark$

Probability Analysis. $\mathcal{B}$ succeeds when:

- (1) Correct identity guess: probability $1/q_{H_1}$.
- (2) Correct forgery index guess: probability $\geq 1/q_{H_2}$.
- (3) No premature abort (guaranteed by weak OMU): probability 1.
- (4) Forking success: probability $\geq \epsilon^2/q_{H_2} - 1/(2q)$.
- (5) Non-zero τ^* : probability $\geq 1 - 1/q$.

Combining:

$$\mathbf{Adv}_{\mathcal{B}}^{q\text{-SDH}}(\lambda) \geq \frac{1}{q_{H_1} q_{H_2}} \left(\frac{\epsilon^2}{q_{H_2}} - \frac{1}{2q} \right).$$

Rearranging and applying $\sqrt{A+B} \leq \sqrt{A} + \sqrt{B}$:

$$\epsilon \leq q_{H_2} \sqrt{q_{H_1} \cdot \mathbf{Adv}_{\mathcal{B}}^{q\text{-SDH}}(\lambda)} + q_{H_2} \sqrt{\frac{3}{2q} + \frac{1}{q_{H_1}}}.$$

This completes the proof. $\square$

5.2 Computational Blindness

Theorem 2 (Computational Blindness in the ROM). *Let H_2 be modeled as a random oracle. Then the scheme achieves computational blindness against any PPT adversary $\mathcal{A}$ playing the role of a malicious signer. Concretely, for any PPT adversary $\mathcal{A}$ making at most q_{H_2} queries to H_2 ,*

$$\mathbf{Adv}_{\mathcal{A}}^{\text{blind}}(\lambda) \leq \frac{2q_{H_2}}{q}.$$

Proof. Let $\mathcal{A}$ be any PPT adversary playing the blindness experiment $\text{Exp}_{\mathcal{A},b}^{\text{blind}}(\lambda)$ with advantage ϵ . We analyze the distribution of the signer's view in the two concurrent sessions.

Signer's Complete View. The signer's view across both sessions is:

$$\text{View} = (w_1^{(0)}, w_{\text{raw}}^{(0)}, h''^{(0)}, w_1^{(1)}, w_{\text{raw}}^{(1)}, h''^{(1)}).$$

For session $j \in \{0, 1\}$:

$$\begin{aligned} w_1^{(j)} &= g^{k_1^{(j)}}, \\ w_{\text{raw}}^{(j)} &= (w_1^{(j)})^{k_2^{(j)}} = g^{k_1^{(j)} k_2^{(j)}}, \\ h''^{(j)} &= (k_1^{(j)} \alpha^{(j)})^{-1} \cdot H_2 \left(M_{b \oplus j} \parallel (w_{\text{raw}}^{(j)})^{\alpha^{(j)}} \right) \pmod{q}, \end{aligned}$$

where $k_1^{(j)}, \alpha^{(j)} \xleftarrow{\$} \mathbb{Z}_q^*$ are chosen uniformly at random by the user and are never transmitted to the signer.

Step 1: $w_1^{(j)}$ and $w_{\text{raw}}^{(j)}$ are message-independent. The values $w_1^{(j)} = g^{k_1^{(j)}}$ and $w_{\text{raw}}^{(j)} = g^{k_1^{(j)} k_2^{(j)}}$ are determined solely by the user's random scalar $k_1^{(j)}$ and the signer's random scalar $k_2^{(j)}$, neither of which depends on the message $M_{b \oplus j}$. Hence these components of the view are identically distributed for $b = 0$ and $b = 1$.

Step 2: The blinded value $\tilde{w}^{(j)}$ is hidden from the signer. Define $\tilde{w}^{(j)} := (w_{\text{raw}}^{(j)})^{\alpha^{(j)}} = g^{k_1^{(j)} k_2^{(j)} \alpha^{(j)}}$. The signer observes $w_{\text{raw}}^{(j)} = g^{k_1^{(j)} k_2^{(j)}}$ but not $\tilde{w}^{(j)}$. Since $\alpha^{(j)} \xleftarrow{\$} \mathbb{Z}_q^*$ is chosen uniformly and independently of everything the signer has seen, and since the map $\alpha \mapsto g^{k_1^{(j)} k_2^{(j)} \alpha}$ is a bijection on $\mathbb{G}_T \setminus \{1\}$ for any fixed nonzero $k_1^{(j)} k_2^{(j)}$, the value $\tilde{w}^{(j)}$ is uniformly distributed over $\mathbb{G}_T \setminus \{1\}$ from the signer's perspective, independently of the message $M_{b \oplus j}$.

Step 3: The signer cannot query H_2 on $\tilde{w}^{(j)}$. Since $\tilde{w}^{(j)}$ is never transmitted to the signer, and since $\tilde{w}^{(j)}$ is uniformly distributed over $\mathbb{G}_T \setminus \{1\}$ (a set of size $q - 1$), the probability that the signer queries $H_2(\cdot \parallel \tilde{w}^{(j)})$ for the correct $\tilde{w}^{(j)}$ in any of its q_{H_2} queries is at most $q_{H_2}/(q - 1)$. By a union bound over both sessions:

$$\Pr \left[\mathcal{A} \text{ queries } H_2(\cdot \parallel \tilde{w}^{(j)}) \text{ for some } j \right] \leq \frac{2q_{H_2}}{q - 1}.$$

Step 4: $h''^{(j)}$ is uniformly distributed and message-independent. Condition on the event $\mathcal{E}$ that $\mathcal{A}$ does not query $H_2(\cdot \parallel \tilde{w}^{(j)})$ for either j . Under $\mathcal{E}$, the value $H_2(M_{b \oplus j} \parallel \tilde{w}^{(j)})$ is uniformly distributed over $\mathbb{Z}_q^*$ from the signer's perspective (since H_2 is a random oracle and $\tilde{w}^{(j)}$ has not been queried). Since $k_1^{(j)} \alpha^{(j)} \neq 0$ in $\mathbb{Z}_q^*$, multiplication by $(k_1^{(j)} \alpha^{(j)})^{-1}$ is a bijection on $\mathbb{Z}_q^*$, so $h''^{(j)}$ is uniformly distributed over $\mathbb{Z}_q^*$, independently of the message $M_{b \oplus j}$.

Step 5: Identical view distributions. Under $\mathcal{E}$, the joint view

$$\text{View} = (w_1^{(0)}, w_{\text{raw}}^{(0)}, h''^{(0)}, w_1^{(1)}, w_{\text{raw}}^{(1)}, h''^{(1)})$$

has identical distribution for $b = 0$ and $b = 1$: each $h''^{(j)}$ is uniform over $\mathbb{Z}_q^*$ and independent of the message assignment. Therefore, conditioned on $\mathcal{E}$, $\mathcal{A}$ has zero advantage:

$$\mathbf{Adv}_{\mathcal{A}}^{\text{blind}}(\lambda) \leq \Pr[\neg \mathcal{E}] \leq \frac{2q_{H_2}}{q-1} \leq \frac{2q_{H_2}}{q}.$$

Since $q \approx 2^{256}$ and q_{H_2} is polynomial in λ , this bound is negligible. $\square$

Remark 2 (Role of the Blinding Factors). The two user-side scalars k_1 and α serve distinct roles:

- (i) k_1 randomises $w_1 = g^{k_1}$, providing the user's contribution to the session randomness and enabling the unblinding step $S = [k_1\alpha]S'$.
- (ii) α blinds the hash input: $\tilde{w} = w_{\text{raw}}^\alpha$ is computationally hidden from the signer in the random oracle model, preventing the signer from computing $H_2(M\|\tilde{w})$ and testing candidate messages.

Together, they ensure that breaking blindness requires the signer to query H_2 on the hidden value $\tilde{w}^{(j)}$, which occurs with negligible probability.

Remark 3 (Comparison with the Previous Design). In a previous version of the protocol, the user sent $h'' = k_1^{-1} \cdot H_2(M\|w_{\text{raw}})$ without the additional blinding factor α . In that design, the signer could test candidate messages M_0, M_1 by computing $w_1^{h''} = g^{H_2(M_b\|w_{\text{raw}})}$ and comparing with $g^{H_2(M_c\|w_{\text{raw}})}$ for $c \in \{0, 1\}$ —a computation requiring only one exponentiation and two hash queries, with no need to solve any hard problem. The introduction of α prevents this attack: the signer would need to compute $\tilde{w} = w_{\text{raw}}^\alpha$, which requires either knowing α (secret) or querying H_2 on $\tilde{w}$, which occurs with negligible probability in the ROM.

6 Performance Comparison

6.1 Notation

Let T_p denote the time for one bilinear pairing, T_e for one exponentiation in $\mathbb{G}_T$, T_{m_1} for one scalar multiplication in $\mathbb{G}_1$, and T_{m_2} for one scalar multiplication in $\mathbb{G}_2$. Based on benchmarks using the PBC library (Type-A curve, 512-bit base field, 80-bit security) on a 2.4 GHz Intel Core i7: $T_p \approx 4.5$ ms, $T_e \approx 1.2$ ms, $T_{m_1} \approx 0.8$ ms, $T_{m_2} \approx 1.0$ ms.

Table 1. Theoretical Complexity Comparison(Signing Phase)

Scheme	Pairings	T_e	T_{m_1}	T_{m_2}	Extra	Security
Zhang–Kim [10]	3	2	2	0	–	Informal
Cao et al. [11]	2	3	2	0	–	Informal
Zhang–Peng [12]	2	2	2	1	–	Informal
Li et al. [14]	3	4	4	2	Key Split	Informal
Zhang et al. [15]	4	5	4	1	FE, MFA	Informal
Ours	$0^\dagger$	3	2	0	–	Proven

[†]One pairing $g = e(P_1, P_{\text{pub}})$ is pre-computed during setup; the online signing cost is zero pairings. FE: Fuzzy Extractor; MFA: Multi-Factor Authentication.

Table 2. Estimated Signing Latency (ms)

Scheme	Pairing Cost	Other Cost	Total (est.)
Zhang–Kim [10]	$3 \times 4.5 = 13.5$	≈ 4.0	≈ 17.5
Cao et al. [11]	$2 \times 4.5 = 9.0$	≈ 5.2	≈ 14.2
Zhang–Peng [12]	$2 \times 4.5 = 9.0$	≈ 4.4	≈ 13.4
Li et al. [14]	$3 \times 4.5 = 13.5$	≈ 8.8	≈ 22.3
Zhang et al. [15]	$4 \times 4.5 = 18.0$	≈ 10.4	≈ 28.4
Ours	0	≈ 5.2	$\approx \mathbf{5.2}$

6.2 Theoretical Complexity Comparison

6.3 Estimated Signing Latency

Our scheme achieves approximately:

- 70% reduction vs. Zhang–Kim [10]
- 63% reduction vs. Cao et al. [11]
- 61% reduction vs. Zhang–Peng [12]
- 77% reduction vs. Li et al. [14]
- 82% reduction vs. Zhang et al. [15]

6.4 Security Comparison

7 Extension: Partial Blind Signatures

Partial blind signatures [4] allow a portion of the message to remain visible to the signer while the rest is blinded. This is useful for applications requiring auditability alongside privacy.

Auditable E-Cash. The bank may require that each coin includes public meta-data (e.g., denomination, expiration date) while the serial number remains blinded.

Table 3. Security Comparison

Scheme	Unforgeability	Blindness	Assumption
Zhang–Kim [10]	Informal	Informal	–
Cao et al. [11]	Informal	Informal	–
Zhang–Peng [12]	Informal	Informal	–
Li et al. [14]	Informal	Informal	–
Zhang et al. [15]	Informal	Informal	–
Ours	WeakOMU (Proven)	Computational (Proven, ROM)	q -SDH + ROM

Conditional Anonymous Credentials. The issuer may require that certain attributes (e.g., “age ≥ 18 ”) are visible during issuance, while other attributes (e.g., exact age, name) remain blinded.

Traceable E-Voting. The election authority may require that each ballot includes public metadata (e.g., election ID, timestamp) while the voter’s choice remains blinded.

7.1 Construction

Message Structure. Let $M = (M_{\text{pub}}, M_{\text{priv}})$ where M_{pub} is public information visible to the signer and M_{priv} is private information blinded from the signer.

Partial Blind Signing Protocol.

1. **Agreement Phase:** User and Signer agree on M_{pub} (transmitted in plaintext).
2. **User $\rightarrow$ Signer:** Choose $k_1 \xleftarrow{\$} \mathbb{Z}_q^*$. Compute $w_1 = g^{k_1} \in \mathbb{G}_T$. Send w_1 .
3. **Signer $\rightarrow$ User:** Choose $k_2 \xleftarrow{\$} \mathbb{Z}_q^*$. Compute $w_{\text{raw}} = w_1^{k_2} \in \mathbb{G}_T$. Send w_{raw} .
4. **User (local blinding):** Choose $\alpha \xleftarrow{\$} \mathbb{Z}_q^*$. Compute $\tilde{w} = w_{\text{raw}}^\alpha \in \mathbb{G}_T$. Compute $h = H_2(M_{\text{pub}} \| M_{\text{priv}} \| \tilde{w}) \pmod{q}$. Compute $h'' = (k_1 \alpha)^{-1} \cdot h \pmod{q}$. Send h'' to Signer.
5. **Signer $\rightarrow$ User:** Compute $l = k_2 - h'' \pmod{q}$. Compute $S' = [l] d_A \in \mathbb{G}_1$. Send S' .
6. **User (local unblinding):** Compute $S = [k_1 \alpha] S' \in \mathbb{G}_1$. Output $\sigma = (h, S)$.

Verification. Given $(A, M = (M_{\text{pub}}, M_{\text{priv}}), \sigma = (h, S))$, verify as a standard SM9signature on the full message $M_{\text{pub}} \| M_{\text{priv}}$.

7.2 Security Properties

Weak OMU. The proof of weak OMU extends naturally: the adversary cannot query signatures on the target identity, and the reduction embeds the q -SDH challenge into the target identity’s private key. The public part M_{pub} does not affect the reduction.

Computational Partial Blindness. The signer learns M_{pub} but learns nothing about M_{priv} beyond what is implied by M_{pub} . The proof of computational blindness (Theorem 2) applies directly to M_{priv} : the signer’s view $(w_1, w_{\text{raw}}, h'')$ satisfies $h'' = (k_1 \alpha)^{-1} \cdot H_2(M_{\text{pub}} \| M_{\text{priv}} \| \tilde{w})$, and since $\tilde{w} = w_{\text{raw}}^\alpha$ is uniformly distributed over $\mathbb{G}_T \setminus \{1\}$ from the signer’s perspective (and never transmitted), h'' is computationally indistinguishable from a uniform element of $\mathbb{Z}_q^*$ independently of M_{priv} . Hence the scheme achieves computational partial blindness with respect to M_{priv} in the random oracle model.

8 Conclusion

We have presented an efficient SM9-based identity-based blind signature scheme with rigorous security proofs. We formalized the weak OMU model and demonstrated its sufficiency for e-cash, anonymous credentials, and e-voting. We proved weak OMU under the standard q -SDH assumption (without Gap assumptions) and *computational blindness* in the random oracle model (without additional computational assumptions). The blinding factor α , introduced to prevent the comparison attack present in earlier constructions, adds only one exponentiation in $\mathbb{G}_T$ to the user’s online cost while providing provable computational blindness in the ROM. Our scheme eliminates all online pairing computations through pre-computation, achieving 61–82% latency reduction compared to existing SM9-based blind signatures. Our signatures are standard SM9 signatures, ensuring interoperability with existing SM9 implementations. We finally extended our scheme to partial blind signatures, enabling auditable e-cash and conditional credentials.

The combination of rigorous security proofs, optimal efficiency, and full standard compatibility makes our construction immediately deployable in systems requiring compliance with Chinese national cryptographic standards.

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